supply-context

File: supply-context.kicad_sch

input-protection

File: input-protection.kicad_sch

v6-regulated-domain

File: v6-regulated-domain.kicad_sch

v6-rgb-translation

File: v6-rgb-translation.kicad_sch

gpio-isolation

Sheet: /
File: v6-protection.kicad_sch

Title: V6 protected power / GPIO circuit study — NOT MAIN

Size: A4	Date:	**Rev: V6 UNQUALIFIED**
KiCad E.D.A. 9.0.2		Id: 6/6

ISOLATED U2 CORRECTION ONLY. Accepted PCB and other V6 nets unchanged.
MAX38902A: SELA=GND, SELB open selects3.3V. R163 >=0.1mA floor requires total resistance error <=1%.
Intrinsic reverse protection; no MODE GPIO. High IQ cost remains in USB suspend budget.



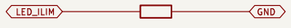
F1
SF-1206F100-2 CANDIDATE



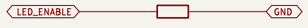
U2
MAX38902AATA+T



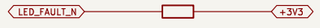
U7
TPS2553DBVR-1 EXISTING



R112
133k1% EXISTING — budget review OPEN



R109
100k EXISTING



R113
10k EXISTING



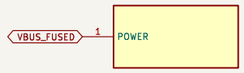
C100
4.7uF 50V X7R 10%



C101
4.7uF 50V X7R 10%



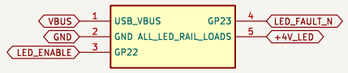
C142
10nF 50V C0G 5%



#FLG01
POWER AFTER PASSIVE FUSE



R163
31.6k 0.1% 10ppm/K



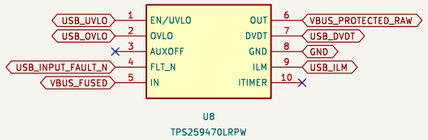
P904
EXTERNAL/MAIN BOUNDARY ONLY

ISOLATED BOM/FOOTPRINT STUDY. No electrical net changes from reviewed correction.
Exact parts and official land pattern now specified; effective C/ESR/hot leakage/thermal gates OPEN.
31.6k load floor requires total resistance error <=1%; see bounded drift budget.
BYP-OUT10nF: ~0.66ms initial ramp, ~330ms short recovery, neither a guaranteed timing bound.
Reset/USB recovery requires hardware rail qualification, not a firmware delay.

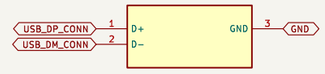
Sheet: /supply-context/
File: supply-context.kicad_sch

Title: U2 correction — NOT FOR FAB

Size: A3	Date:	**Rev: STUDY ONLY**
KiCad E.D.A. 9.0.2		Id: 2/6



U8
TPS259470LRPW



U10
TPD2EUSB30DRTR



R120
590k 0.1%

R121
156k 0.1%



R122
620k 0.1%

R123
267k 0.1%



C125
1uF 50V effective TBD



C126
1uF 50V



R124
6.65k 0.1%



C124
10nF 10% 50V



R126
100k 1%



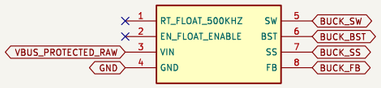
P901
EXISTING BOARD INTERFACE

Sheet: /input-protection/
File: input-protection.kicad_sch

Title: Maru60 input protection candidate — fixed direct USB-C

Size: A3	Date:	**Rev: ISOLATED STUDY 4 — NO**
KiCad E.D.A. 9.0.2		Id: 3/6

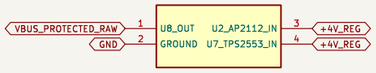
V6 CIRCUIT STUDY ONLY: replaces V4 LT1963 stage. No PCB/main adoption.
U8/fuse input protection and GPIO sheet remain separate V4 interfaces.
30V buck input rating removes dependency on typical OVLO delay for U2/U7 INPUT ratings.
It does NOT guarantee output transient, USB current, startup or suspend compliance.



U18
TPS62932DRLR



L2
XAL4030-472MEC 4.7uH 20%



P906
NONPHYSICAL BOUNDARY



R159
40.0k 0.1%



R160
10.0k 0.1%



C136
100nF 50V



C137
4.7uF 50V effective>=2uF TBD



C138
100nF 16V BST-SW



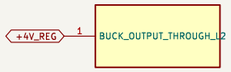
C139
100nF 10% 16V - soft start



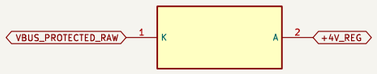
C140
22uF 10V effective>=12uF TBD



C141
22uF 10V effective>=12uF TBD



P908
REGULATOR OUTPUT FLAG



D100
PMEG4030ER - reverse discharge bypass

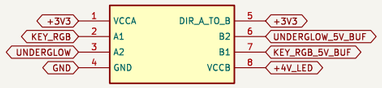
RT intentionally floating: normal switching frequency450..550kHz. Dropout/PFM folding separate.
FB nominal4.0V; static reference/resistor extrema3.9137..4.0865V, excluding FB bias/ripple/transients.
C140/C141 exact MPN, effective capacitance and compensation review OPEN.
U8 downstream raw-domain capacitors must be >=35V; C124 remains50V.
No direct tie of EN to raw VIN: EN absolute maximum6V.
C199 is reserved for separate ADC_AVDD work; no reference collision.

Sheet: /v6-regulated-domain/
File: v6-regulated-domain.kicad_sch

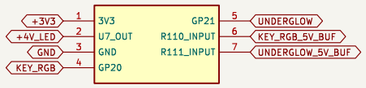
Title: Maru60 V6 regulated domain - ISOLATED UNQUALIFIED CIRCUIT

Size: A3	Date:	**Rev: ISOLATED STUDY 1 - NO**
KiCad E.D.A. 9.0.2		Id: 4/6

Replaces U5/U6 AHCT buffers, NOT a third stream or GPIO allocation change.
VCCA=3.3V; VCCB=post-U7 LED rail. Preserves two independent PIO data streams.
No claim of arbitrary partially powered/brownout hot-plug tolerance.



U5
SN74LVC2T45DCUR



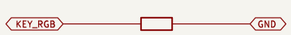
P907
NONPHYSICAL BOUNDARY



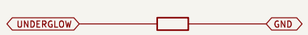
C122
100nF 16V



C123
100nF 16V



R161
47.0k 1% - reset data LOW



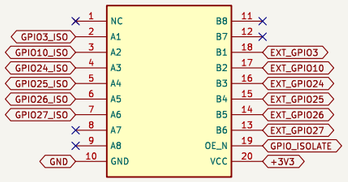
R162
47.0k 1% - reset data LOW

Ioff is specified at one supply exactly0V; brownout transitions remain a defined test/review gate.
Firmware holds both data inputs LOW before LED enable/disable.
LED part minimum3.7V: prove worst-case far-end rail, not only regulator nominal.
The downstream TPS2553 + routing drop, current policy and cold-start sequence remain OPEN.

Sheet: /v6-rgb-translation/
File: v6-rgb-translation.kicad_sch

Title: Maru60 V6 RGB dual supply interface - isolated circuit

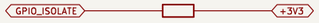
Size: A3	Date:	**Rev: ISOLATED STUDY 1 - NO**
KiCad E.D.A. 9.0.2		Id: 5/6



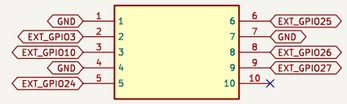
U11
SN74CB3T3245PWR



U12
TLV810EA29DBZR



R128
100k 1%



J3
SM10B-SRSS-TB



C128
100nF 16V



C129
100nF 16V



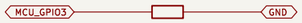
R130
1.00k 1%



R131
1.00k 1%



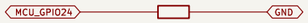
R132
1.00k 1%



R140
39.0k 1%



R141
39.0k 1%



R142
39.0k 1%



R133
1.00k 1%



R134
1.00k 1%



R135
1.00k 1%



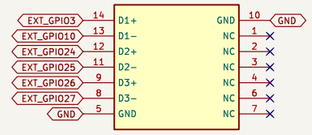
R143
39.0k 1%



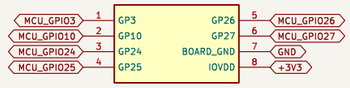
R144
39.0k 1%



R145
39.0k 1%



U13
TPD6E05U06RVZR



~~P902~~
EXISTING MCU INTERFACE

Sheet: /gpio-isolation/
File: gpio-isolation.kicad_sch

Title: *Maru60 protected digital-only six GPIO header candidate*

Size: A3	Date:	**Rev: ISOLATED STUDY 4 – NO**
KiCad E.D.A. 9.0.2		Id: 6/6